Jeremiah Dibble

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SUMMARY

Senior Machine Learning Engineer (4+ years) specializing in the **full lifecycle of Large Language Models (LLMs)**. Expertise spans **model pre-training, synthetic SFT pipelines, and preference tuning**. Skilled in designing, building, and deploying production-ready software workflows powered by LLMs, while leveraging business context to improve efficiency and/or performance.

EDUCATION

Georgia Institute of Technology – MS, Computer Science (in progress)
California Institute of Technology – BS, Electrical Engineering & Business, Economics, and Management (2021)

EXPERIENCE

Nuclearn – Full-time

Acting Infrastructure Manager — **August 2025 - Present**

- Led the transition to **high availability hosting** of the web platform and custom built LLMs using Ceph on Proxmox and Litellm as a load balancer for the LLMs.
- Led scoping and purchasing **GPU hosting hardware** for expanding hosting training both in house and for customers' datacenters.
- Setup **distributed training across multiple GPU servers** from hardware purchase all the way to SLURM and Mellanox configuration.

Senior Machine Learning Engineer — **June 2024 - Present**

- Designed and built a **high-throughput VLM-based OCR** service. Architected the system as Dockerized FastAPI endpoints with a Redis task queue and distributed workers, dispatching page-level inference to remote vLLM servers and opportunistically leveraging idle development GPUs to scale capacity.
- Workers **dynamically route inference** to the least-loaded vLLM server based on real-time GPU metrics, achieving **~20 pages/second per 8-GPU node** and thousands of pages per minute when scaling across multiple servers.
- Created a **self-aligning synthetic data pipeline**, based on novel research, to create synthetic LLM training data from proprietary customer data.
- Using this method I created hundreds of thousands of domain specific conversations and improved domain specific instruction following score for our different models by 20%-55%, based on internal benchmarking tools.
- **Pre- and post-trained domain specific LLMs on billions of tokens** to improve the model's understanding of nuclear topics and its ability to generate nuclear specific documents and answers.
- Filtered and **judged** customer conversations to iteratively improve model performance.
- Implemented a **comparative preference optimization** generation and training to allow for tuning of models towards superior outputs, which resulted in up to a 13% improvement on internal benchmarking.

VoxQI - Qualitative Intelligence – Full-time — September 2023 – June 2024

Lead Machine Learning Engineer

- Designed, implemented, and deployed a Retrieval, Augmentation, and Generation **(RAG) chatbot** and associated custom semantic search vector database to allow customers to extract insights from previous campaigns and user feedback.
- Developed an API to search, cluster, and organize extensive user feedback using RAG, enabling the generation of tailored reports and summaries. Successfully deployed in Docker on AWS Lambda.

Worthix – Full-time — December 2022 - July 2023

Machine Learning Engineer

- Spearheaded the development of a proof-of-concept for **using LLMs in primary survey analysis**. Tested and implemented internal LLMs and workflows, **resulting in a 25% improvement** in completeness, accuracy, and specificity of survey analysis.
- Presented technical performance and product design changes of the LLM approach to leverage or mitigate its strengths and weaknesses. Collaborated with the team to design and implement a **REST API** for testing the new **LLM analysis** and product design with customers.
- Played a key role in testing and designing a new analysis workflow. This workflow utilized multistage models to predict and recommend during surveys, ultimately increasing feedback specificity and accuracy.

Caltech's Office of Technology Transfer and Corporate Partnerships – Full-time — June 2021 - December 2022

Machine Learning Engineer

- Implemented and tested **custom loss functions** from research papers to combine transformer models to add an unlimited number of additional tasks or contexts with significantly less additional training than traditional training. This led to **50%+ savings on compute costs** due to shorter training times.
- Improved **model sparsity algorithm** using **Pytorch** to be compatible with a more general set of NLP model architectures and automated the sparsification at a variety of different levels based on optimizing for a given target sparsity. Researchers in the lab leveraged these tools for data collection used to publish papers.
- Automated **supervised and unsupervised fine-tuning of BERT or GPT NLP models** on any given text dataset, to allow for the mass fine-tuning of different models and datasets.
- Preprocessed **Waymo traffic point cloud** into different formats to optimize the **self-organizing neural net's (SNN)** predictions of motion and interaction of multiple objects to determine the market viability of the SNN.